Resource Asset Registration Guide
V5.5 11/01/19

Revision History

Date	Version	Description	Author
10/10/2007	0.03	Draft for Internal Review	D. Showalter
10/10/2007	0.04	Second draft for Internal Review	D. Showalter
10/10/2007	0.05	Draft for Posting	D. Showalter
10/11/2007	0.06	Revised Draft for Posting	D. Showalter
10/16/2007	0.07	Revised with Market Comments	D. Showalter
12/13/2007	0.08	Revised for Planning Submittal	D. Showalter
4/01/2008	3.99	Draft –Reorganized and reformatted for RARF Ver 4	D. Showalter
4/08/2008	4.00	Released with RARF Ver 4 (Official RARF)	D. Showalter
12/16/08	4.01	Updated RARF Guide V4.01	A. Moy
2/4/2009	4.02	Updated and re-wrote transmission and load data tabs	S. Middleton
3/10/2009	4.03	Corrected / Modified business rules for transformer tab	S. Middleton
6/5/2009	4.04	Added Business Rules for Planning / Protection	S. Middleton
6/10/2009	4.05	Added Business Rules for Planning / Protection	P. Nellutla
6/23/2009	4.06	Added Business Rules for Planning / Protection	P. Nellutla
07/06/09	4.07	Updated Business Rules for Planning, Line Data, Transformer tab and updated the definition of unit commercial date in section 4, Date effective field is included in section 12-18	P. Nellutla
07/24/09	4.08	Updated Business Rules for Line Data, Transformer, Breaker/Switch, Series Device, Load Data, Capacitor Reactor, Static Var Compensator, Planning, Protection. Updated Section 7.4	P. Nellutla
09/01/09	4.09	Updated Section 7.3 and updated Business Rules for Transformer/Line/Breaker-Switch/Series Device Data/Capacitor or Reactor/Load Data.	P. Nellutla
11/2/2009	4.10	Updated Business Rules for Capacitor/Reactor and Transformer tabs.	P. Nellutla
11/18/2009	4.11	Updated section 6.5 with Business Rules for Operational Parameters-GEN, CC, and Wind. Updated Business Rules for Capacitor- Reactor Tab	P. Nellutla
2/8/2010	4.12	Updated Business Rules for Operational/Transmission tabs	P. Nellutla
12/10/10	4.13	Added Protocol Disclaimer	A. Deller
6/16/2011	4.14	Removed TML and added MIS link	G.Schroeder
10/26/11	5.00	Updated Guide for RARF v5.0	A. Deller
12/08/11	5.00	Updated Guide for RARF v5.0	A. Moy
3/20/12	5.00	Updated Guide for RARF v5.0	A. Moy
06/18/2012	5.1	Updated Guide for RARF v5.1	G. Schroeder
12/09/2013	5.2	Updated Guide for RARF v5.2 Wind Form for NPRR425	G. Schroeder
11/17/2014	5.3	Updated Guide for RARF v5.3 NER Regional Standard BAL-TRE-001	G.Schroeder
5/5/17	5.4	Update Guide for RARF v5.4	A.Deller
11/01/19	5.5	Update Guide for v5.5	G.Schroeder

Table of Contents

PROTOCOL DISCLAIMER

This Guide describes ERCOT processes and is not intended to be a substitute for the ERCOT Nodal Protocols (available at http://www.ercot.com/mktrules/nprotocols/), as amended from time to time. If any conflict exists between this document and the ERCOT Nodal Protocols, the ERCOT Nodal Protocols shall control in all respects.

1.0 Summary of Resource Registration Guide

This document is a guide to completing Resource Asset Registration Form (RARF) with ERCOT in accordance with Section 16 of the ERCOT Nodal Protocols. Upon obtaining the forms from Resource Entities, ERCOT will keep the RARFs in a central repository hub so the files can be tracked and easily accessed by all ERCOT systems, as well as communicated back to the Resource Entity through audits (Figure 1 below illustrates the process flow of receiving and loading RARF data).

Data from the RARF package is used to provide input to the Network Model used by Operations in dispatch of Resources. RARF data also feeds the Planning group to support system studies. Data also flows to the Registration systems as well as the Settlement and Aggregation systems to ensure proper settlement of Resource dispatches and operation.

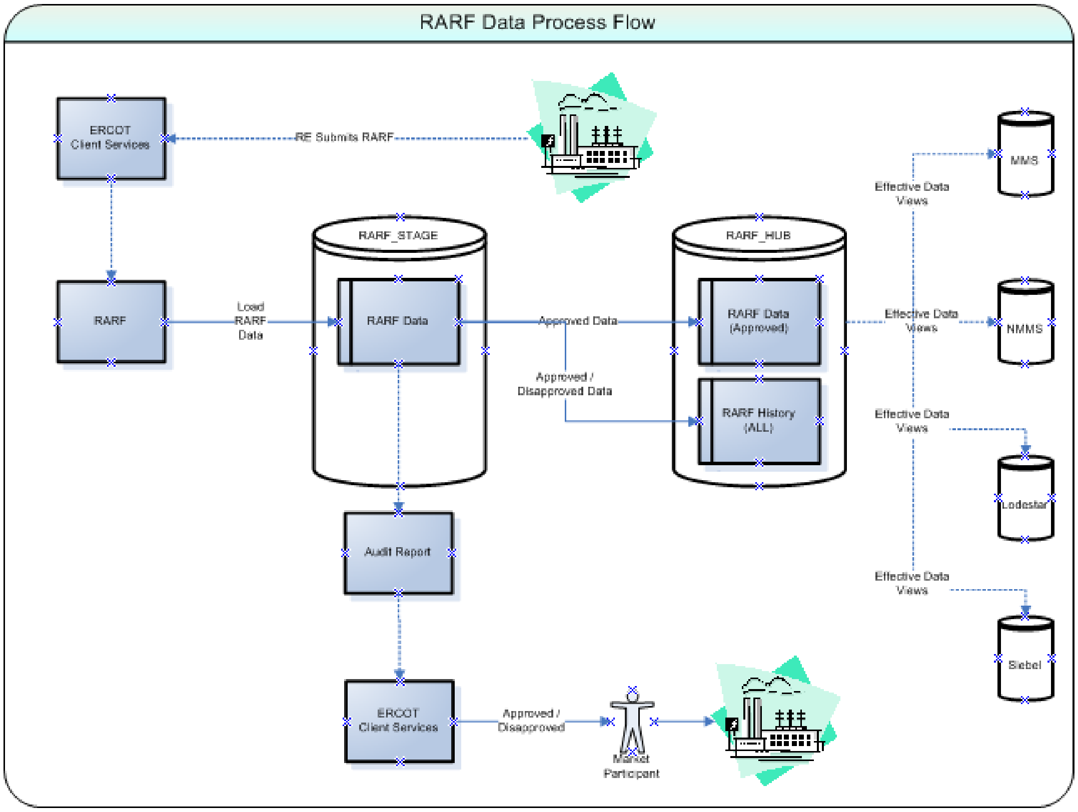


Figure 1

2.0 Summary of Resource Asset Registration Forms

2.1 Tabs

The RARF is a collection of workbooks that focus on specific Resource types. There are 7 major workbooks, as follows:

- General Site ESIID Information
- Generation
- Combined Cycle
- Renewable
- Load Resource
- Transmission
- Settlement Only Generation (SOG) Gen Form Pivoted

Each of the workbooks has multiple tabs for organizing and collecting data as follows:

- General Site ESIID Information
    - Instructions
    - General and Site Information
    - GEN Load Split – ESIID
    - Private Network - Site

- Generation
    - Instructions
    - Unit Info – GEN
    - Unit Info - AGR
    - Ownership – GEN
    - Parameters – GEN
    - Operational Parameters – Gen
    - Operational Parameters – NRRC
    - Operational Parameters – ERRC
    - Private Network – Unit
    - Reactive Capability – GEN
    - Planning – GEN
    - Protection – GEN
    - SubSync – GEN
    - PSCAD Model
    - Dynamic Data

- Renewable
    - Instructions
    - Unit Info – RENEWABLE
    - Wind Turbine details
    - Inverter Details – SOLAR
    - Panel Details – SOLAR
    - Ownership - RENEWABLE
    - Parameters – RENEWABLE
    - Operational Parameters – RENEWABLE
    - Operational Parameters – NRRC
    - Operational Parameters – ERRC
    - Private Network - Unit
    - Reactive Capability – RENEWABLE
    - Planning – RENEWABLE

- Protection – RENEWABLE
    - Collector System - RENEWABLE
    - Collector System – Renew Segment Data
    - PSCAD Model
    - Dynamic Data

- Combined Cycle
    - Instructions
    - Unit Info – TRAIN
    - Unit Info - CC
    - Ownership – CC
    - Parameters – CC
    - CC Configurations
    - CC Transitions
    - Parameters – CFG
    - Operational Parameters – CFG
    - Operational Parameters – NRRC
    - Operational Parameters – ERRC
    - Private Network - Unit
    - Reactive Capability – CC
    - Planning – CC
    - Protection – C
    - Subsync - CC
    - PSCAD Model
    - Dynamic Data

- LaaR
    - RARF Form
    - General Information – LAAR
    - Load Resource Information
    - Load resource Parameters
    - CLR-NRRC
    - CLR-ERRC

- Transmission
    - Instructions
    - Station
    - Line Data
    - Line Temperature
    - Most Limiting Series Element
    - Breaker Switch Data
    - Capacitor and Reactor Data
    - Transformer Data
    - Transformer Tap Settings
    - Static Var Compensator Data
    - Series Device Data
    - Load Data
    - PUN Load
    - One Line
    - Transformer Test Data

- Settlement Only Generation (SOG)
    - Instructions
    - Unit Info - DG

2.2 Glossary

The RARF Glossary is posted on the ERCOT website (http://www.ercot.com/mktrules/guides/resourcereg/library) and is controlled by the Resource Data Working Group. This document controls what data elements are in the RARF and also defines what data is required at each stage of registration.

3.0 Instructions

A RARF package should be submitted for each generation resource site that contains data for all generation at the site. A separate RARF should also be submitted for each Resource Entity covering all load resources represented by that entity. A RARF is to be completed for all active and mothballed generation resources inside ERCOT. Organizations must submit a market participant application as a Resource Entity prior to submission of this form, if not eligible for Federal Hydro waiver (Section 16.5). If questions arise related to the completion of this form, please contact your designated ERCOT Account Manager or email Model Administration at resourcereg@ercot.com with the subject "Resource/Asset Registration Form".

- Please bear in mind the following for the completion of this form, A single, complete Generation RARF package, consists of:
    - the General Site ESIID Information workbook,
    - the Generation, Combined Cycle, Wind and/or Settlement Only Generation (SOG) workbook(s) as appropriate and
    - the workbook for Transmission assets, must be submitted for each resource site.
    - Note (In each form on the second tab in cell B5 is a cell that references the Site Code, this MUST be completed on all 5 of the (Gen, CC, Renewable, Settlement Only Generation (SOG) and Transmission) forms.)
- A single RARF must be submitted for each load resources represented by a common Resource Entity.

3.1 Process for Official Submittal

Each submittal must have all three workbooks (General Site ESIID Information the Resource workbook(s) (Generation, Combined Cycle, Renewable, Settlement Only Generation (SOG)) and the Transmission workbook). Even if the submission is to alter a generator parameter, the General Site ESIID Information, the appropriate resource workbook (Generation, Combined Cycle, Renewable and/or Settlement Only Generation (SOG)) and Transmission book must be included. The LAAR will have a single workbook.

There are two methods of submitting the RARF, as follows:

PRIMARY: RARFs are to be submitted through the Market Information System (MIS) located at https://mis.ercot.com. Submission through the MIS link requires a valid Authorized Representative's digital certificate.

ALTERNATE: An alternate email signature document is available upon request from your ERCOT Account Manager for those who have technical problems submitting via MIS. The RARF must be emailed in Microsoft Excel spreadsheet (xls) format, along with the signature document to: mpappl@ercot.com and resourcereg@ercot.com . (Please allow at least 2 additional days for processing)

The following are instructions for submitting the RARF through MIS:

- Access to ERCOT MIS requires a user digital certificate with a minimal role of "MP_ASSESTS' this allows access to "Create Service Request" on the "Applications" page. The "user digital certificate" is authorized by the Market Participant's User Security Administrator.

- Upon accessing MIS, click on "*Applications*" then select "*Service Request*". Be advised that the Service Request will display in a new window as a pop-up, which may be restricted by browser settings.

- Complete the required fields on the "*Service Request*" screen annotated by red asterisks.

- The following Request Type and Sub-Type are essential to a proper submittal:

  - *Request Type:* Select "**MP Registration**" from the drop-down list
  - *Request Sub-Type:* Select "**Resource/Asset Registration**" from the drop-down list

> *Please note that if the Type and Sub-Type values above are not used, the RARF will not be received or processed by ERCOT Client Services.*

  - Short Description: Brief description of change and Sitecode.

  - Long Description: List of changes that are being made to the RARF.
    Example: Parameters, Operational Parameters, Reactive Capabilities, Breaker Switch, etc.

- Click "*Submit*" (you will add the RARF file on the next screen)

- From the "*Activities and Attachments*" screen, under the Attachments heading of the Service Request click the 'Add' button. Please make sure that your Attachment is a .ZIP file and all RARF forms that are included are in .XLS format.

> *Please note that the RARF Forms must keep version 5.0 naming convention when being submitted (Wind_Form_Pivoted.xls, Gen_Form_Pivoted.xls, Combined_Cycle_Form_Pivoted.xls and Transmission_Form_Pivoted.xls).*
>
> *The LAAR Form must keep the naming convention LAAR_Form_Pivoted.xls.*

- Click "*Submit*" (you will add the RARF Zip file on the next screen)

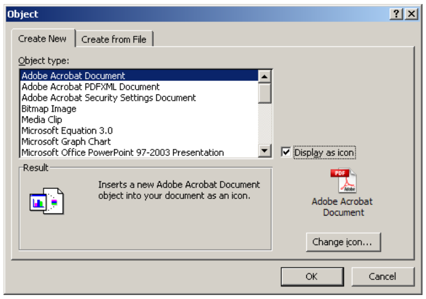
- Click "*Submit*" (comments are optional)

ERCOT will verify the RARF is sent from the Authorized Representative of the registered Resource Entity via digital certificate. ERCOT may request additional authentication as deemed necessary.

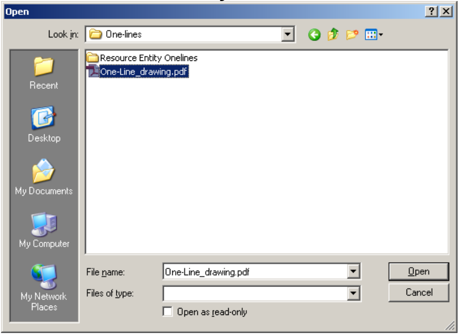
3.2 Process for Embedding documents in the RARF

1. While on the tab needing to embed a document within, click on INSERT, then click on OBJECT

2. From the popup screen click on Adobe Acrobat Document in the list and also check the 'display as icon' button, then OK.



3. A new popup screen will appear to select the file you want to embed, and then click the 'Open' button.



NOTE: the one-line pdf file will Open into a new screen. Just close it down and the file will be embedded in the One Line tab.

4.0 General Information and ESIID Information RARF Workbook

4.1 Instructions

Detailed instructions for the RARF submittal process through the Market Information System (MIS) are contained in section 3.1 of this document.

4.2 General and Site Information

The General and Site Information tab identifies information about the Resource Entity. The contact information is essential, as it provides ERCOT with additional contacts in case of questions regarding the RARF. The Resource Entity Name does not allow any special characters except spaces and dashes. The Data Universal Numbering System (DUNS) number is a nine digit identifier, some companies are given an additional 4 digits (total13 digits) by ERCOT to uniquely identify their entity. The Resource site name and site code are unique for each site and the code is jointly coordinated with ERCOT. The Site Name and Site Code should follow the name on the Interconnect Agreement signed with the Transmission Service Provider, with the Site Code being no more than 8 characters and all CAPs, no spaces or special characters (except underscores).

	A	B
1	*ERCOT Confidential*	
2	**General Information - All Resource Entities**	
3	*This worksheet tab contains information on the Resource Entity responsible for submitting this form.*	
4	*Please complete this section.*	
5		
6		
7	**This submittal is for:**	
8	** Deletions are accepted as intentions. This form does not supersede the Notice of Suspension of Operations requirements.*	
9		
10	**Submittal Information**	
11	Date Form Completed:	
12	Resource Entity Submitting Form:	
13	Resource Entity DUNS #:	
14		
15	**Site Information**	
16	Resource Site Name:	
17	Resource Site Code:	
18	Street Address:	
19	City:	
20	State	
21	Zipcode	
22	County:	
23	Site In-Service Date:	
24	Site Stop Service Date:	
25	Congestion Management Zone for 2003:	
26	Resource owned by NOIE?	
27	Is Resource behind a NOIE Settlement Meter Point?	
28	Number of EPS Primary meters:	
29	Is Resource claiming status as a Settlement Only Generator (SOG) as defined in ERCOT Protocol Section 2.1, Definitions?	
30	Is Resource >10 MW?	
31		
32	**Primary Contact (name of person ERCOT can contact with questions regarding this form)**	
33	Printed Name:	
34	Title:	
35	Phone Number:	
36	E-mail Address:	
38		
39	**Secondary Contact (if available)**	
40	Printed Name:	
41	Title:	
42	Phone Number:	
43	E-mail Address:	
45		

4.3 Generation Load Split Information

This section must be completed for the Electric Service Identifier assigned to a service delivery point. Or to note if the facility has or does not have a Gen Site Load split among multiple competitive retailers or among multiple TDSPs or has multiple Electric Service Identifiers.

With Split Load Resources

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GENERATION SITE LOAD SPLITTING DETAILS

Complete this section for ESI ID information and Gen Site Load Details

GENERATION SITE LOAD DETAILS							
Generation Load Split?	ERCOT Read Meter?	ESI ID:	TDSP Providing Service To Resource:	TDSP DUNS Number:	Fixed Load Splitting	Load Serving Entity	Load Serving Entity DUNS #
Y/N	Y/N		List	Automatic		List	Automatic
Y	Y	111111111999	TDSP ONE	123456789	0.5	LSE ONE	999999999
		111111111888	TDSP TWO	123456788	0.5	LSETWO	999999990

Without Split Load Resources

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GENERATION SITE LOAD SPLITTING DETAILS

Complete this section for ESI ID information and Gen Site Load Details

GENERATION SITE LOAD DETAILS							
Generation Load Split?	ERCOT Read Meter?	ESI ID:	TDSP Providing Service To Resource:	TDSP DUNS Number:	Fixed Load Splitting	Load Serving Entity	Load Serving Entity DUNS #
Y/N	Y/N		List	Automatic		List	Automatic
N	N	111111111999		123456789	1		

4.4 Private Network - Site

The Private Network Site tab identifies whether the site is part of a private use network (PUN) which typically contains load that is not directly metered by ERCOT. If the site is part of a PUN then the data is required for the site Load characteristics for the MW load and the MVAR load percentages. If the site is not part of a PUN, then no other data need be filled out on this tab and additional fields are blacked out.

With Private Network Site Info

SITE DETAILS	PRIVATE NETWORK - SITE INFORMATION							Load Characteristics Fc		
Private Network?	Average Amount Of Self-serve Private Load	Average Amount Of Self-serve Private Reactive Load	Expected Typical Private Network Net Interchange	Expected Typical Private Network Net Reactive Interchange	Private Network Gross Unit Capability	Private Network Gross Unit Reactive Capability	Large Motor, Percent Of Total MW Load	Small Motor, Percent Of Total MW Load	Resistive (heat Load, Percent Total MW Lo
Y/N	MW	MVAR	MW	MVAR	MW	MVAR	%	%	%
Y	50.00	10.00	75.00	40.00	90.00	42.00	75.00%	20.00%	2.00%

Without Private Network Site Info

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Private Network - Site Information

This worksheet tab applies to ***all Private Use Networks.***

Must complete all cells in this section if this is a Private Network Site.

SITE DETAILS	PRIVATE NETWORK - SITE INFORMATION						Load Characteristics For MW Load					Load Characteristics For MVAR Load			
Private Network?	Average Amount Of Self-serve Private Load	Average Amount Of Self-serve Private Reactive Load	Expected Typical Private Network Net Interchange	Expected Typical Private Network Net Reactive	Private Network Gross Unit Capability	Private Network Gross Unit Reactive Capability	Large Motor, Percent Of Total MW Load	Small Motor, Percent Of Total MW Load	Resistive (heating) Load, Percent Of Total MW Load	Discharge Lighting, Percent Of Total MW Load	Other, Percent Of Total MW Load	Large Motor, Percent Of Total MVAR Load	Small Motor, Percent Of Total MVAR Load	Discharge Lighting, Percent Of Total MVAR Load	Other, Percent Of Total MVAR Load
Y/N	MW	MVAR	MW	MVAR	MW	MVAR									
N															

5.0 Generation

5.1 Instructions

Detailed instructions for the RARF submittal process through the Market Information System (MIS) are contained in section 3.1 of this document.

5.2 Unit Info - GEN

This section must be completed for each unit that is part of the generation resource site. It includes the Generator details, site and unit dates, point of interconnection information, fuel information, ratings and design information and geographic location data. (Fields include: Settlement Only Generation (SOG), Transmission POI date, SGIA date, Model Ready date, POI substation, POI voltage, POI bus number, Primary fuel, Secondary fuel, Resource category, Physical unit type, Governor droop, Governor dead-band, Design Max ambient temp, Design Min ambient temp, Latitude of plant & Longitude of plant). Note: Cell B5 (Resource Site Code) must be completed on all forms.

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Unit Information

This worksheet tab provides generator Unit information for ***Generation*** *resources.*

Must complete all applicable cells in this section.

Resource Site Code:

		GENERATOR DETAILS					SITE AND UNIT	
UNIT NAME	Resource Name (Unit Code/Mnemonic)	Settlement Only Generator (SOG)	PUC Registration Number	ERCOT Interconnection Project Number - Only New Units	NERC Number	Qualifying Facility	Transmission Only MRD	Standard Generation Interconnection Agreement (SGIA) Signature Date
All Caps	Automatic	Y/N				Y/N	mm/dd/yyyy	mm/dd/yyyy

Note: Prior to your next RARF submission, remove all unit(s) that are no longer active and were stopped in a prior submission.

5.3 Unit Info – AGR

This section must be completed for each unit that is part of the generation with an Aggregated Unit . It includes the Resource Name Unit Code/Mnemonic, Aggregated Generation Resource Manufacturer Model, MW Rating for this Aggregated Generation Resource and Number of this type of Aggregated Generation Resource.

*Note – even if a site doesn't have an AGR, the unit name still is required to be filled in.

A	B	C	D
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Aggregated Generation Resource Information			
*This worksheet tab provides **Aggregated Generation Resources** information for generation resources.*			
Must complete all applicable cells in this section.			
	AGR DETAILS		
Resource Name (Unit Code/Mnemonic)	Aggregated Generation Resource(Manufacturer/Model)	MW Rating for this Aggregated Generation Resource	Number of this type of Aggregated Generation Resource
Automatic	all caps	MW	#

5.4 Ownership - GEN

The Ownership tab must be completed for each unit that is part of the generation resource site. It includes the Generator details, ownership information and percentages. (Fields include: Unit Name, Resource Name, Ownership start date & Ownership stop date)

A	B	C	D	E	F	G	H
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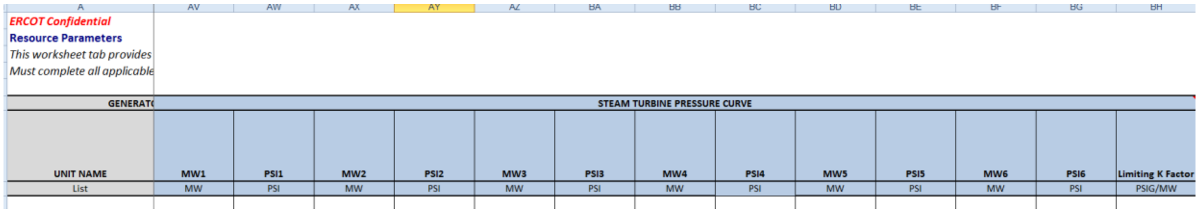
Representation of Facility Output							
This worksheet tab applies to all Generation Resources. This tab identifies the Resource Entities that represent the output of the facility to ERCOT.							
Must complete all applicable cells in this section.							
GENERATOR DETAILS		OWNERSHIP PERCENTAGES					
Unit Name	Resource Name (Unit Code/Mnemonic)	Joint Ownership?	Resource Entity Name	Resource Duns Number	Fixed Ownership	Master Owner	Ownership Start Date
List	List	Y/N		9-13 digits		Y/N	mm/dd/yyyy

5.5 Parameters - GEN

The Parameters tab must be completed for each unit that is part of the generation resource site. It includes the Generator details, reasonability limits, seasonal ratings and unit de-rating values for temperature changes. (Fields include: Unit Name, Resource Name & Unit De-rating values)

GENERATOR DETAILS		REASONABILITY LIMITS				SEASONAL RATINGS - SPRING			
UNIT NAME	Resource Name (Unit Code/Mnemonic)	High Reasonability Limit	Low Reasonability Limit	High Reasonability Ramp Rate Limit	Low Reasonability Ramp Rate Limit	Seasonal Net Max Sustainable Rating - Spring	Seasonal Net Min Sustainable Rating - Spring	Seasonal Net Max Emergency Rating - Spring	Seasonal Net Min Emergency Rating - Spring
List	Automatic	MW	MW	MW/min	MW/min	MW	MW	MW	MW
UNITA	TEST_UNITA	100.00	20.00	5.00	1.00	75.00	50.00	100.00	20.00

The Parameters tab has been updated to include Steam Turbine Pressure Curve data for Generation units with Physical Unit Type (ST) and Resource Category is 'Gas Steam - Non-reheat or Boiler without air-preheater, Gas Steam - Reheat Boiler, Gas Steam - Supercritical Boiler or Coal and Lignite'.



ERCOT Confidential
Resource Parameters
This worksheet tab provides
Must complete all applicable

A	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH
GENERATO	STEAM TURBINE PRESSURE CURVE												
UNIT NAME	MW1	PSI1	MW2	PSI2	MW3	PSI3	MW4	PSI4	MW5	PSI5	MW6	PSI6	Limiting K Factor
List	MW	PSI	MW	PSI	MW	PSI	MW	PSI	MW	PSI	MW	PSI	PSIG/MW

Field NAME	Definition / Detailed Description	Validations
MW1	Maximum pressure. Net MW value where the steam generator typically reaches rated pressure. If pressure is constant for the normal operating range enter the same value as is entered as the Governor minimum operating level (required value for steam turbines).	1. MW1 Must Not be Null if Physical Unit is 'ST' and Resource Category is 'Gas Steam - Non-reheat or Boiler without air-preheater, Gas Steam - Reheat Boiler, Gas Steam - Supercritical Boiler or Coal and Lignite' . 2. MW1 must be a floating point. 3. MWH => MW1 => MWL where, MWH = HRL MWL = Max(MW2*, MW3*, MW4*, MW5*, MW6, LRL) ((* do not include in computation if null))
PSI1	Rated throttle pressure (required value for steam turbines) at MW1	1. PSI1 must be a floating point. 2. If MW1 is non-null then PSI1 must be non-null.
MW2	Net unit output (breakpoint value used to define the pressure/MW curve). If not needed, enter same values as the Pressure PSI (1) or MW (1).	1. MW2 must be a floating point. 2. If PSI2 is non-null then MW2 must be non-null. 3. If MW2 is non-null, MWH => MW2 => MWL where, MWH = Min (HRL, MW1) MWL = Max(MW3*, MW4*, MW5*, MW6, LRL) ((* do not include in computation if null))
PSI2	Throttle steam pressure (PSI) at MW2 value (breakpoint value used to define the pressure/MW curve). If not needed, enter same values as the Pressure PSI (1) or MW (1).	1. PSI2 must be a floating point. 2. If PSI2 is non-null then MW2 must be non-null.
MW3	Net unit output (breakpoint value used to define the pressure/MW curve). If not needed, enter same values as the Pressure PSI (1) or MW (1).	1. MW3 must be a floating point. 2. If PSI3 is non-null then MW3 must be non-null. 3. If MW3 is non-null, MWH => MW3 => MWL where, MWH = Min (HRL, MW1, MW2*) ((* do not include in computation if null)) MWL = Max(MW4*, MW5*, MW6, LRL) ((* do not include in computation if null))
PSI3	Throttle steam pressure (PSI) at MW3 value (breakpoint value used to define the pressure/MW curve). If not needed, enter same values as the Pressure PSI (1) or MW (1).	1. PSI3 must be a floating point. 2. If MW3 is non-null then PSI3 must be non-null.
MW4	Net unit output (breakpoint value used to define the pressure/MW curve). If not needed, enter same values as the Pressure PSI (1) or MW (1).	1. MW4 must be a floating point. 2. If PSI4 is non-null then MW4 must be non-null. 3. If MW4 is non-null, MWH => MW4 => MWL where, MWH = Min (HRL, MW1, MW2*, MW3*) ((* do not include in computation if null)) MWL = Max(MW5*, MW6, LRL) ((* do not include in computation if null))
PSI4	Throttle steam pressure (PSI) at MW4 value (breakpoint value used to define the pressure/MW curve). If not needed, enter same values as the Pressure PSI (1) or MW (1).	1. PSI4 must be a floating point. 2. If MW4 is non-null then PSI4 must be non-null.
MW5	Net unit output (breakpoint value used to define the pressure/MW curve). In not needed, enter same values as the Pressure PSI (1) and MW (1).	1. MW5 must be a floating point. 2. If PSI5 is non-null then MW5 must be non-null. 3. If MW5 is non-null, MWH => MW5 => MWL where, MWH = Min (HRL, MW1, MW2*, MW3*, MW4*) ((* do not include in computation if null)) MWL = Max(MW6, LRL)
PSI5	Throttle steam pressure (PSI) at MW5 value (breakpoint value used to define the pressure/MW curve). If not needed, enter same values as the Pressure PSI (1) or MW (1).	1. PSI5 must be a floating point. 2. If MW5 is non-null then PSI5 must be non-null.
MW6	Minimum pressure. Net unit MW output where the steam generator typically reaches minimum pressure (required value for steam turbines)	1. MW6 Must Not Be Null if Physical Unit is 'ST' and Resource Category is 'Gas Steam - Non-reheat or Boiler without air-preheater, Gas Steam - Reheat Boiler, Gas Steam - Supercritical Boiler or Coal and Lignite' . 2. MW6 must be a floating point. 3. MWH => MW6 => MWL where, MWH = Min (HRL, MW1, MW2*, MW3*, MW4*, MW5*) ((* do not include in computation if null)) MWL = LRL

PSI6	Throttle steam pressure (PSI) at MW6 value (required value for steam turbines).	1. PSI6 must be a floating point. 2. If MW6 is non-null then PSI6 must be non-null.
Limiting K Factor	The K factor is used to model the stored energy available to the resource. The value ranges between 0.0 and 0.6 PSIG per MW change when responding during a FME. The resource can measure the drop in throttle pressure when the resource is operating near 50% output of the steam turbine during a FME and provide this ratio of pressure change to the BA. The value is given as a PSI Change per MW change. The default value would be zero (required for steam turbines).	1.K_FACTOR Must Not be Null if Physical Unit is 'ST' and Resource Category is 'Gas Steam - Non-reheat or Boiler without air-preheater, Gas Steam - Reheat Boiler, Gas Steam - Supercritical Boiler or Coal and Lignite' . 2. K Factor must be a floating point with default 0.0 3. 0.0 =< K =< 0.6

5.6 Operational Parameters - GEN

The Operational Parameters tab must be completed for each unit that is part of the generation resource site. It includes the Generator details and the resource parameters including minimum on-line and off-line times, hot, cold, intermediate, hot to intermediate and intermediate to cold start times and daily and weekly start data.

GENERATOR DETAILS		RESOURCE PARAMETERS								
Unit Name	Resource Name (Unit Code/Mnemonic)	Minimum On Line Time	Minimum Off Line Time	Hot Start Time	Intermediate Start Time	Cold Start Time	Max Weekly Starts	Max On Line Time	Max Daily Starts	Max Weekly Energy
List	Automatic	hours	hours	hours	hours	hours		hours		MWh
UNITA	TEST_UNITA	20.00	1.00	2.00	4.00	5.00	24	20.00	3	1000

5.7 Operational Parameters - NRRC

The Operational Parameters Normal Ramp Rate Curve data must be completed for each unit that is part of the generation resource site. Note that only 1 MW value and the corresponding normal upward and normal downward ramp rate are required, additional MW’s are optional.

GENERATOR DETAILS		NORMAL RAMP RATE CURVE				
Unit Name	Resource Name (Unit Code/Mnemonic)	MW Number	NRRC Code	NRRC MW	Upward Ramprate	Downward Ramprate
List	Automatic	List	Automatic	MW	MW/min	MW/min
UNIT1	TEST_UNIT1	MW1	TEST_UNIT1_MW1_NRRC	50.00	5.00	5.00
UNIT1	TEST_UNIT1	MW2	TEST_UNIT1_MW2_NRRC	70.00	7.00	5.00
UNIT1	TEST_UNIT1	MW3	TEST_UNIT1_MW3_NRRC	100.00	10.00	5.00

5.8 Operational Parameters - ERRC

The Operational Parameters Emergency Ramp Rate Curve data must be completed for each unit that is part of the generation resource site. Note that only 1 MW value and the corresponding normal upward and normal downward ramp rate are required, additional MW’s are optional.

GENERATOR DETAILS		EMERGENCY RAMP RATE CURVE				
Unit Name	Resource Name (Unit Code/Mnemonic)	MW Number	ERRC Code	ERRC MW	Upward Ramprate	Downward Ramprate
List	Automatic	List	Automatic	MW	MW/min	MW/min
UNIT1	TEST_UNIT1	MW1	TEST_UNIT1_MW1_ERRC	50.00	7.00	5.00
UNIT1	TEST_UNIT1	MW2	TEST_UNIT1_MW2_ERRC	75.00	10.00	7.00
UNIT1	TEST_UNIT1	MW3	TEST_UNIT1_MW3_ERRC	110.00	15.00	7.00

5.9 Private Network - Unit

The Private Network Unit tab is to be completed only if the site has completed the site information as a PUN site, then the data must be completed for each unit that is part of the private network resource site, Which includes the MW and MVAR load, net interchange and gross unit capabilities. If the site is not part of a PUN, then no unit data need be filled out on this tab and all fields will be blacked out. (Fields include: Unit Name, Site Code & Resource Name)

GENERATOR DETAILS			PRIVATE NETWORK - UNIT INFORMATION							
Unit Name	SITE CODE	Resource Name (Unit Code/Mnemonic)	Average Amount Of Self-serve Private Load	Average Amount Of Self-serve Private Reactive Load	Expected Typical Private Network Net Interchange	Expected Typical Private Network Net Reactive Interchange	Private Network Gross Unit Capability	Private Network Gross Unit Reactive Capability	If Unit Trips, Does Load Trip?	If Yes, Approximate Percentage Of Load That Will Trip?
List	Automatic	Automatic	MW	MVAR	MW	MVAR	MW	MVAR	Y/N	%
UNITA	TEST	TEST_UNITA	50.00	10.00	75.00	40.00	90.00	42.00	N	

5.10 Reactive Capability - GEN

The Reactive Capability must be completed for each unit that is part of the generation resource site. It includes the generator details and the reactive capability curve data which are 5 increasing MW values of Operating Real Power and 9 points of MVAR reactive power information. The lagging MVAR amounts should be listed as positive numbers and the leading MVAR amounts should be listed as negative. MW5 is the unity power factor amount or maximum value of real power MW at 1.0 PF, the reactive power which crosses the x-axis on the curve is zero. (Fields include: Unit Name, Resource Name, Gross/Net values, NDCRC test & Test date)

GENERATOR DETAILS		RE/								
Unit Name	Resource Name (Unit Code/Mnemonic)	Reactive Capability Data Provided is Gross/Net Values?	Reactive Capability Data Provided is from NDCRC Test Data?	If Reactive Capability Data Provided Is From NDCRC test Data Then Enter The Date On Which The Test Was Performed.	Mw1	Lagging MVAR Limit Associated With Mw1 Output	Leading MVAR Limit Associated With Mw1 Output	Mw2	Lagging MVAR Limit Associated With Mw2 Output	Leading MVAR Lim Associated With Mw2 Output
List	Automatic	List	Y/N	mm/dd/yyyy	MW	MVAR	MVAR	MW	MVAR	MVAR
UNITA	TEST_UNITA	GROSS	Y	01/01/2011	50.00	70.00	-69.00	75.00	72.00	-65

5.11 Planning - GEN

The Planning tab must be completed for each unit that is part of the generation resource site. It includes the generator details, planning details, generator auxiliary load information, generation auxiliary MW load characteristics and generation auxiliary MVAR load characteristics. (Fields include: Unit Name, Resource Name, Direct Axis Sub transient reactance unsaturated, Direct Axis Transient reactance unsaturated, Positive Sequence Z – R unsaturated, Positive Sequence Z – X unsaturated, Negative Sequence Z – R unsaturated, Negative Sequence Z – X unsaturated, Zero Sequence exist, Zero Sequence Z – R unsaturated, Zero Sequence Z – X unsaturated, Zero Sequence Grounding Resistance, Zero Sequence Grounding Reactance, Zero Sequence Resistance, Zero Sequence Reactance & Aux Load power factor)

GENERATOR DETAILS											
Unit Name	Resource Name (Unit Code/Mnemonic)	What Is The Mva Base That The Following Data Is Based On?	What Is The Kv Base That The Following Data Is Based On?	Direct Axis Subtransient Reactance, X"di (unsaturated)	Direct Axis Transient Reactance, X'di (unsaturated)	Positive Sequence Z (unsaturated)		Negative Sequence Z (unsaturated)		Does Zero Sequence Z exist?	Zero Seq (unsatu
List	Automatic	MVA	kV	pu	pu	R in p.u.	X in p.u.	R in p.u.	X in p.u.	Y/N	R in p.u.
UNITA	TEST_UNITA	100.00	18.00	0.15000	0.20000	0.00250	0.21000	0.01800	0.18900	Y	0.00150

5.12 Protection - GEN

The Protection tab must be completed for each unit that is part of the generation resource site. It includes the generator details, plant under and over voltage protection and plant under and over frequency protection information. (Fields include: Unit Name & Resource Name)

GENERATOR DETAILS									
Unit Name	Resource Name (Unit Code/Mnemonic)	Breaker Interruption Time	Instantaneous Undervoltage Trip	Undervoltage 1	Time 1	Undervoltage 2	Time 2	Undervoltage 3	
List	Automatic	cycles	p.u.	p.u.	sec	p.u.	sec	p.u.	
UNITA	ST_UNITA	3		7.10000	2.10	5.50000	1.50		

5.13 Subsync - GEN

The Subsync tab should be completed for each unit that may require subsynchronous resonance studies as part of the interconnection process or for additional planning studies to maintain the reliability of the ERCOT System by meeting all NERC Reliability Standards, ERCOT Protocols and Operating Guides that would be affected by the interconnection and operation of the generation site. (Fields include: Unit Name, Resource Name, Mass # & Mass Code)

GENERATOR DETAILS		SUBSYNCHRONOUS RESONANCE						
Unit Name	Resource Name (Unit Code/Mnemonic)	Mass #	Mass Code	Name	Mass Inertia	Inertia Units	Associated Damping	Damping Units
List	Automatic	List	Automatic					
UNITA	TEST_UNITA	MASS1	TEST_UNITA_MASS1	Turbine	6546654.000	lb-ft	6.000	seconds
UNITA	TEST_UNITA	MASS2	TEST_UNITA_MASS2	Exciter	321654.000	lb-ft	5.000	seconds

5.14 PSCAD Model

Embed PSCAD model to this tab. See section 3.2 for process.

5.15 Dynamic Data

Embed Dynamic Data model to this tab. See section 3.2 for process.

6.0 Combined Cycle RARF Workbook

6.1 Instructions

Detailed instructions for the RARF submittal process through the Market Information System (MIS) are contained in section 3.1 of this document.

6.2 Unit Info - TRAIN

This section must be completed for each combined cycle train that is part of the generation resource site. A Train is defined as: The combinations of gas turbines and steam turbines in an electric generation plant that employs more than one thermodynamic cycle. It includes the CC Train details, site and train unit dates, fuel information, power augmentation and geographic location data. (Fields include: Settlement Only Generation (SOG), Resource category, Latitude of plant & Longitude of plant). Each train code is made up of the Site Code appended with a train code (e.g. CC1, CC2 etc.). Note: Cell B5 (Resource Site Code) must be completed on all forms.

	A	B	C	D	E	F	G	H
1	***ERCOT Confidential***							
2	**Unit Information**							
3	*This worksheet tab applies to all **Combined Cycle Train** generation resources.*							
4	*Must complete all applicable cells in this section.*							
5	**Resource Site Code:**							
6	**CC TRAIN DETAILS**						**SITE AND UNIT DATES**	
7	**Train Name**	**Train Code**	**Settlement Only Generator (SOG)**	**PUC Registration Number**	**NERC Number**	**Qualifying Facility?**	**Train Commercial Operations Date**	**Train Retirement Date**
8	**List**	**Automatic**	**Y/N**			**Y/N**	**mm/dd/yyyy**	**mm/dd/yyyy**
9								

Note: Prior to your next RARF submission, remove all Train(s) that are no longer active and were stopped in a prior submission.

6.3 Unit Info - CC

This section must be completed for each unit that is part of the generation resource site. It includes the generator details, site and unit dates, point of interconnection information, fuel information and ratings and design information. (Fields include: Site Code, Train Code, Settlement Only Generation (SOG), Transmission POI date, SGIA date, Model Ready date, POI Substation, POI voltage, POI bus number, Primary fuel, Secondary fuel, Physical unit type, Governor droop, Governor dead-band, Design Max ambient temp & Design Min ambient temp)

GENERATOR DETAILS						SITE AND UNIT DATES		
UNIT NAME	SITE_CODE	Resource Name (Unit Code/Mnemonic)	Train Name	Non_Modeled Generator	ERCOT Interconnection Project Number - Only New Units	Transmission Only Date (POI)	Standard Generation Interconnection Agreement (SGIA) Signature Date	Unit Start Date (Model Ready Date)
all caps	Automatic	Automatic	List	Y/N		mm/dd/yyyy	mm/dd/yyyy	mm/dd/yyyy
UNIT1	TEST	TEST_UNIT1	CC1	N		05/01/2000	05/01/2000	08/01/2000
UNIT2	TEST	TEST_UNIT2	CC1	N		05/01/2000	05/01/2000	08/01/2000
ST1	TEST	TEST_ST1	CC1	N		05/01/2000	05/01/2000	08/01/2000

Note: Prior to your next RARF submission, remove all unit(s) that are no longer active and were stopped in a prior submission.

6.4 Ownership - CC

The Ownership tab must be completed for each train that is part of the generation resource site. It includes the train details and ownership details for each train. (Fields include: Train Name, Train Code, Ownership Start date & Ownership Stop date). Ownership for CC Trains is restricted to one owner each.

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Representation of Facility Output

This worksheet tab applies to all Generation Resources. This tab identifies the Resource Entities that represent the output of the facility to ERCOT.

Must complete all applicable cells in this section.

A	B	C	D	E	F	G	H
GENERATOR DETAILS		OWNERSHIP PERCENTAGES					
Unit Name	Resource Name (Unit Code/Mnemonic)	Joint Ownership?	Resource Entity Name	Resource Duns Number	Fixed Ownership	Master Owner	Ownership Start Date
List	List	Y/N		9-13 digits		Y/N	mm/dd/yyyy

6.5 Parameters - CC

The Parameters CC tab must be completed for each unit that is part of the generation resource site. It includes the Generator details, reasonability limits, seasonal ratings and unit de-rating values for temperature changes. (Fields include: Unit Name, Resource Name & Unit de-rating values)

GENERATOR DETAILS		REASONABILITY LIMITS				SEASONAL RATINGS - SPRING				
Unit Name	Resource Name (Unit Code/Mnemonic)	High Reasonability Limit	Low Reasonability Limit	High Reasonability Ramp Rate Limit	Low Reasonability Ramp Rate Limit	Seasonal Net Max Sustainable Rating - Spring	Seasonal Net Min Sustainable Rating - Spring	Seasonal Net Max Emergency Rating - Spring	Seasonal Net Min Emergency Rating - Spring	Seasonal Net Max Sustainable Rating - Summer
List	Automatic	MW	MW	MW/min	MW/min	MW	MW	MW	MW	MW
UNIT1	TEST_UNIT1	160.00	100.00	12.00	2.00	157.00	105.00	157.00	105.00	154.0
UNIT2	TEST_UNIT2	176.00	100.00	12.00	2.00	157.00	105.00	157.00	105.00	153.0
ST1	TEST_ST1	200.00	100.00	10.00	2.00	180.00	100.00	200.00	100.00	180.0

6.6 CC Configuration

The Combined Cycle Configurations tab must be completed for each train that is part of the generation resource site. It includes the Train and Generator details and the operational configurations for each operationally unique train. (Fields include: Train Code, Site Code Resource Name, Configuration # & Configuration Code). Configurations are identified by number. The configuration code is automatically generated by concatenating the Train Code and the Configuration Code. “x” indicates a primary units for the configuration, “A” indicates an alternate unit for a configuration.

GENERATOR DETAILS			CONFIGURATION DETAILS		
TRAIN CODE	SITE CODE	Resource Name (Unit Code/Mnemonic)	Configuration #	Configuration Code	Configuration Type
List	Automatic	List	#	Automatic	List
TEST_CC1	TEST	TEST_UNIT1	1	TEST_CC1_1	X
TEST_CC1	TEST	TEST_UNIT2	1	TEST_CC1_1	A
TEST_CC1	TEST	TEST_UNIT1	2	TEST_CC1_2	X
TEST_CC1	TEST	TEST_UNIT2	2	TEST_CC1_2	A
TEST_CC1	TEST	TEST_ST1	2	TEST_CC1_2	X

Note Regarding Alternate Units in CC Configurations:

If the units G1, G2, G3 all have the same unit characteristics such that the three configurations CC1_2, CC1_3, CC1_4 all have the same resource capabilities and would be offered exactly the same way, the 'A' CT can be the alternate for any of the 3 CTs, and there would not be a need to register CC1_3 or CC1_4 in the example below. That is to say, if CC1_2 was the only registered 2x1 configuration, it could be used to represent any combination of 2 CTs running with the CA unit. For example, if one of the G1, G2, or G3 units were in outage, you could use CC1_2 to run the remaining 2 non-outaged CTs running with the CA.

Resource Name (Unit Code)	Unit Type	TEST_CC1_1	TEST_CC1_2	TEST_CC1_3	TEST_CC1_4	TEST_CC1_5	TEST_CC1_6
TEST_G1	CT	X	X	X	A	X	X
TEST_G2	CT	X	X	A	X	X	X
TEST_G3	CT	X	A	X	X	X	A
TEST_G5	CA	X	X	X	X		

6.7 CC Transitions

The Combined Cycle Transitions tab must be completed for each train that is part of the generation resource site. It includes the Train details and transitions for each train. The transitions are a map that, for each operating state/configuration that you are in, it identifies what states/configurations you can go to next – whether it is adding a unit or removing a unit. Note you must complete the configurations tab before data can be entered on this tab. (Fields include: Site Code, Train Code, Configuration Code To & Configuration Code From). Each configuration must be able to transition from a configuration of "OFFLINE" to at least one valid configuration and from at least one valid configuration to "OFFLINE".

TRAIN DETAILS		TRANSITIONS	
Sitecode	Mnemonic for Combined Cycle Train	Configuration Code From	Configuration Code To
List	List	List	List
TEST	TEST_CC1	OFFLINE	TEST_CC1_1
TEST	TEST_CC1	TEST_CC1_1	TEST_CC1_2
TEST	TEST_CC1	TEST_CC1_2	TEST_CC1_1
TEST	TEST_CC1	TEST_CC1_1	OFFLINE

6.8 Parameters - CFG

The Parameters CFG tab must be completed for each configuration that is part of a train. It includes the Generator details, reasonability limits, seasonal ratings and unit de-rating values for temperature changes for each configuration. (Fields include: Site Code, Train Code, Configuration Code & Unit de-rating values)

TRAIN DETAILS			REASONABILITY LIMITS				SEASONAL RATINGS - SPRING				
SITECODE	Train Code	Configuration Code	High Reasonability Limit	Low Reasonability Limit	High Reasonability Ramp Rate Limit	Low Reasonability Ramp Rate Limit	Seasonal Net Max Sustainable Rating - Spring	Seasonal Net Min Sustainable Rating - Spring	Seasonal Net Max Emergency Rating - Spring	Seasonal Net Min Emergency Rating - Spring	Sea Max S Ratin
List	List	List	MW	MW	MW/min	MW/min	MW	MW	MW	MW	
TEST	TEST_CC1	TEST_CC1_1	160.00	100.00	12.00	2.00	157.00	105.00	157.00	105.00	
TEST	TEST_CC1	TEST_CC1_2	360.00	200.00	22.00	4.00	337.00	205.00	357.00	205.00	

6.9 Operational Parameters - CFG

The Operational Parameters CFG tab must be completed for each configuration that is part of a train. It includes the Generator details and the resource parameters including minimum on-line and off-line times, hot, cold, intermediate, hot to intermediate and intermediate to cold start times and daily and weekly start data. Note you must complete the configurations tab before data can be entered on this tab. (Fields include: Site Code, Train Code & Configuration Code)

TRAIN DETAILS			RESOURCE PARAMETERS									
SITECODE	Train Code	Configuration Code	Minimum On Line Time	Minimum Off Line Time	Hot Start Time	Intermediate Start Time	Cold Start Time	Max Weekly Starts	Max On Line Time	Max Daily Starts	Max Weekly Energy	Hot-to-intermediate Time
LIST	LIST	LIST	hours	hours	hours	hours	hours				MWh	hours
TEST	TEST_CC1	TEST_CC1_1	2.00	2.00	3.00	4.00	6.00	7	6000.00	1	25200	1.00
TEST	TEST_CC1	TEST_CC1_2	6.00	2.00	3.00	4.00	6.00	7	6000.00	1	39000	1.00

6.10 Operational Parameters - NRRC

The Operational Parameters Normal Ramp Rate Curve data must be completed for each configuration that is part of a train. Only 1 MW value and the corresponding normal upward and normal downward ramp rate are required, additional MW's are optional. Note you must complete the configurations tab before data can be entered on this tab. (Fields include: Train Code, Configuration Code & MW number)

		NORMAL RAMP RATE CURVE				
Train Code	Configuration Code	MW Number	NRRC Code	NRRC MW	Upward Ramprate	Downward Ramprate
List	List	List	Automatic	MW	MW/min	MW/min
TEST_CC1	TEST_CC1_1	MW1	TEST_CC1_1_MW1_NRRC	100.00	20.00	10.00
TEST_CC1	TEST_CC1_1	MW2	TEST_CC1_1_MW2_NRRC	150.00	25.00	10.00
TEST_CC1	TEST_CC1_1	MW3	TEST_CC1_1_MW3_NRRC	200.00	40.00	10.00

6.11 Operational Parameters - ERRC

The Operational Parameters Emergency Ramp Rate Curve data must be completed for each configuration that is part of a train. Only 1 MW value and the corresponding normal upward and normal downward ramp rate are required, additional MW's are optional. Note you must complete the configurations tab before data can be entered on this tab. (Fields include: Train Code, Configuration Code & MW number)

		EMERGENCY RAMP RATE CURVE				
Train Code	Configuration Code	MW Number	ERRC Code	ERRC MW	Upward Ramprate	Downward Ramprate
LIST	LIST	List	Automatic	MW	MW/min	MW/min
TEST_CC1	TEST_CC1_1	MW1	TEST_CC1_1_MW1_ERRC	100.00	20.00	15.00
TEST_CC1	TEST_CC1_1	MW2	TEST_CC1_1_MW2_ERRC	150.00	25.00	15.00
TEST_CC1	TEST_CC1_1	MW3	TEST_CC1_1_MW3_ERRC	200.00	40.00	15.00

6.12 Private Network - Unit

The Private Network Unit tab is to be completed only if the site has completed the site information as a PUN site, then the data must be completed for each unit that is part of the private network resource site, Which includes the MW and MVAR load, net interchange and gross unit capabilities. If the site is not part of a PUN, then no unit data need be filled out on this tab and all fields will be blacked out. (Fields include: Unit Name, Site Code & Resource Name)

GENERATOR DETAILS			PRIVATE NETWORK - UNIT INFORMATION						
Unit Name	SITE CODE	Resource Name (Unit Code/Mnemonic)	Average Amount Of Self-serve Private Load	Average Amount Of Self-serve Private Reactive Load	Expected Typical Private Network Net Interchange	Expected Typical Private Network Net Reactive Interchange	Private Network Gross Unit Capability	Private Network Gross Unit Reactive Capability	If Unit Trips, Does Load Trip?
List	Automatic	Automatic	MW	MVAR	MW	MVAR	MW	MVAR	Y/N
UNIT1	TEST	TEST_UNIT1	40.00	10.00	40.00	10.00	160.00	50.00	Y
UNIT2	TEST	TEST_UNIT2	40.00	10.00	40.00	10.00	176.00	50.00	Y
ST1	TEST	TEST_ST1	20.00	20.00	20.00	20.00	200.00	40.00	N

6.13 Reactive Capability - CC

The Reactive Capability must be completed for each unit that is part of the generation resource site. It includes the generator details and the reactive capability curve data which are 5 increasing MW values of Operating Real Power and 9 points of MVAR reactive power information. The lagging MVAR amounts should be listed as positive numbers and the leading MVAR amounts should be listed as negative. MW5 is the unity power factor amount or maximum value of real power MW at 1.0 PF, the reactive power which crosses the x-axis on the curve is zero. (Fields include: Unit Name, Resource Name, Gross/Net values, NDCRC test & Test date)

GENERATOR DETAILS										
Unit Name	SITE_CODE	Resource Name (Unit Code/Mnemonic)	Reactive Capability Data Provided is Gross/Net Values?	Reactive Capability Data Provided is from NDCRC Test Data?	If Reactive Capability Data Provided Is From NDCRC test Data Then Enter The Date On Which The Test Was Performed.	Mw1	Lagging MVAR Limit Associated With Mw1 Output	Leading MVAR Limit Associated With Mw1 Output	Mw2	Lagging MVAR Limit Associated With Mw2 Output
List	Automatic	Automatic	List	Y/N	mm/dd/yyyy	MW	MVAR	MVAR	MW	MVAR
UNIT1	TEST	TEST_UNIT1	NET	Y	12/12/1999	100.00	50.00	-50.00	135.00	50.00
UNIT2	TEST	TEST_UNIT2	NET	Y	12/12/1999	100.00	50.00	-50.00	135.00	50.00
ST1	TEST	TEST_ST1	NET	Y	12/12/1999	100.00	50.00	-40.00	135.00	40.00

6.14 Planning - CC

The Planning tab must be completed for each unit that is part of the generation resource site. It includes the generator details, planning details, generator auxiliary load information, generation auxiliary MW load characteristics and generation auxiliary MVAR load characteristics. (Fields include: Unit Name, Site Code, Resource Name, Direct Axis Sub transient reactance unsaturated, Direct Axis Transient reactance unsaturated, Positive Sequence Z – R unsaturated, Positive Sequence Z – X unsaturated, Negative Sequence Z – R unsaturated, Negative Sequence Z – X unsaturated, Zero Sequence exist, Zero Sequence Z – R unsaturated, Zero Sequence Z – X unsaturated, Zero Sequence Grounding Resistance, Zero Sequence Grounding Reactance, Zero Sequence Resistance, Zero Sequence Reactance & Aux Load power factor)

GENERATOR DETAILS									
Unit Name	SITE_CODE	Resource Name (Unit Code/mnemonic)	What Is The Mva Base That The Following Data Is Based On?	What Is The Kv Base That The Following Data Is Based On?	Direct Axis Subtransient Reactance, X"di (unsaturated)	Direct Axis Transient Reactance, X'di (unsaturated)	Positive Sequence Z (unsaturated)		Negative S (unsatu
List	Automatic	Automatic	MVA	kV	pu	pu	R in p.u.	X in p.u.	R in p.u.
UNIT1	TEST	TEST_UNIT1	100.00	18.00	0.15000	0.20000	0.01800	0.18000	0.01700
UNIT2	TEST	TEST_UNIT2	100.00	18.00	0.15000	0.21000	0.01800	0.18500	0.01700
ST1	TEST	TEST_ST1	200.00	24.00	0.19000	0.18000	0.01500	0.15600	0.01400

6.15 Protection - CC

The Protection tab must be completed for each unit that is part of the generation resource site. It includes the generator details, plant under and over voltage protection and plant under and over frequency protection information. (Fields include: Unit Name, Site Code & Resource Name)

GENERATOR DETAILS									
Unit Name	SITE_CODE	Resource Name (Unit Code/mnemonic)	Breaker Interruption Time	Instantaneous Undervoltage Trip	Undervoltage 1	Time 1	Undervoltage 2	Time 2	Undervoltage 3
List	Automatic	Automatic	cycles	p.u.	p.u.	sec	p.u.	sec	p.u.
UNIT1	TEST	TEST_UNIT1	2		0.70000	4.00			
UNIT2	TEST	TEST_UNIT2	2		0.70000	4.00			
ST1	TEST	TEST_ST1	2		0.90000	1.00			

6.16 Subsync - CC

The Subsync tab should be completed for each unit that may require subsynchronous resonance studies as part of the interconnection process or for additional planning studies to maintain the reliability of the ERCOT System by meeting all NERC Reliability Standards, ERCOT Protocols and Operating Guides that would be affected by the interconnection and operation of the generation site. (Fields include: Unit Name, Resource Name, Mass # & Mass Code)

GENERATOR DETAILS			SUBSYNCHRONOUS RESONANCE							
Unit Name	SITE_CODE	Resource Name (Unit Code/mnemonic)	Mass #	Mass Code	Mass Name	Mass Inertia	Inertia Units	Associated Damping	Damping Units	Stiffness Between Masses Previous And Current Mass
List	Automatic	Automatic	List	Automatic						
UNIT1	TEST	TEST_UNIT1	MASS1	TEST_UNIT1_MASS1	Generator Body	2563.000	lbm-ft 2	NA	126470000.000	NA
UNIT2	TEST	TEST_UNIT2	MASS1	TEST_UNIT2_MASS1	Generator Body	2563.000	lbm-ft 2	NA	126470000.000	NA

6.17 PSCAD Model

Embed PSCAD model to this tab. See section 3.2 for process.

6.18 Dynamic Data

Embed Dynamic Data model to this tab. See section 3.2 for process.

7.0 Renewable RARF Workbook

7.1 Instructions

Detailed instructions for the RARF submittal process through the Market Information System (MIS) are contained in section 3.1 of this document.

7.2 Unit Info - RENEWABLE

This section must be completed for each unit that is part of the renewable resource site. It includes the Generator details, site and unit dates, point of interconnection information, fuel information, ratings and design information and geographic location data. (Fields include: Settlement Only Generation (SOG), Transmission POI date, SGIA date, Model Ready date, POI substation, POI voltage, POI bus number, Primary fuel, Secondary fuel, Resource category, Physical unit type, Governor droop, Governor dead-band, Design Max ambient temp, Design Min ambient temp, Height of Instrumentation, Latitude of plant & Longitude of plant). Note: Cell B5 (Resource Site Code) must be completed on all forms.

	A	B	C	D	E	F	G	H	I
1	*ERCOT Confidential*								
2	**Unit Information**								
3	*This worksheet tab applies only to **Renewable** generation resources and should list aggregated WGR and PVGR whether consisting of single or multiple turbine/inverter types.*								
4	*Must complete all applicable cells in this section.*								
5	**Resource Site Code:**								
6			GENERATOR DETAILS						
7	UNIT NAME	Resource Name (Unit Code/Mnemonic)	Settlement Only Generator (SOG)	PUC Registration Number	ERCOT Interconnection Project Number - only new units	NERC Number	Qualifying Facility	Transmission Only MRD	Standard Generation Interconnection Agreement (SGIA) Signature Date
8	All Caps	Automated	Y/N				Y/N	mm/dd/yyyy	mm/dd/yyyy
9									

> *Note: Prior to your next RARF submission, remove all unit(s) that are no longer active and were stopped in a prior submission.*

7.3 Wind Turbine Details

- This section must be completed for each unit and turbine type that makes up the wind resource site. It includes the Wind Generator Group details, Turbine Details, Planning Details and Turbine Step-Up Transformer Data. Fields include: WGR Group #, Turbine Type, Instantaneous Controlled Fault Current Magnitude (Multiple of full load current) for Turbine Types 3 & 4, Controlled Fault Current Magnitude at 2 to 3 cycles after fault (Multiple of full load current) for Turbine Types 3 & 4 and Controlled Fault Current Magnitude at 4 plus cycles after fault (Multiple of full load current) for Turbine Types 3 & 4)

- **A Resource Entity may aggregate wind turbines together to form a WGR subject to provisions of Protocol Section 3.10.7.2 (9).**

- **A Resource Entity may also group WGRs into a WGR Group.** The group is a collection of two or more WGRs whose performance in responding to SCED Dispatch Instructions will be assessed as an aggregate for Generation Resource Energy Deployment Performance

(GREDP). A WGR Group cannot contain any WGRs that are Split Generation Resources. Additionally, only WGRs that have the same Resource Node can be mapped to a WGR Group.

Example: Non-Grouped WGRs

For Non_Grouped WGRs do not complete the WGR Group #.

WIND GENERATION GROUP DETAILS				TURBINE DETAILS			
Resource Name (Unit Code/Mnemonic)	WGR Group	Group Code (Hidden Field on Form)	SITE_GROUP	Turbine Manufacturer/Model	MW Rating for this model of Turbine	Number of Turbine Manufacturer/Model	Turbine Type
List	#	Automatic	Automatic		MW	#	1, 2,3,4,5
SITE_UNIT1		SITE_UNIT1_VESTAS NM72		VESTAS NM72	1.65	25	1
SITE_UNIT2		SITE_UNIT2_VESTAS V80		VESTAS V80	1.8	25	2

Example: Grouped WGRs

Grouped WGRs must complete the WGR Group #.

WIND GENERATION GROUP DETAILS				TURBINE DETAILS			
Resource Name (Unit Code/Mnemonic)	WGR Group	Group Code (Hidden Field on Form)	SITE_GROUP	Turbine Manufacturer/Model	MW Rating for this model of Turbine	Number of Turbine Manufacturer/Model	Turbine Type
List	#	Automatic	Automatic		MW	#	1, 2,3,4,5
SITE_UNIT3	1	SITE_UNIT3_GAMESA G80_1	SITE_1	GAMESA G80	2	25	3
SITE_UNIT3	1	SITE_UNIT3_GAMESA G83_1	SITE_1	GAMESA G83	2	25	3
SITE_UNIT3	1	SITE_UNIT3_GAMESA G80_1	SITE_1	GAMESA G80	2	25	3
SITE_UNIT3A	1	SITE_UNIT3A_GAMESA G87_1	SITE_1	GAMESA G87	2	25	3
SITE_UNIT3B	1	SITE_UNIT3B_GAMESA G87_1	SITE_1	GAMESA G87	2	25	3
SITE_UNIT4	2	SITE_UNIT4_SIEMENS_2	SITE_2	SIEMENS	2.3	25	4
SITE_UNIT4	2	SITE_UNIT4_SIEMENS_2	SITE_2	SIEMENS	2.3	25	4
SITE_UNIT4	2	SITE_UNIT4_SIEMENS_2	SITE_2	SIEMENS	2.3	25	4

7.4 Inverter Details – SOLAR

- This section must be completed for each unit that is part of a solar unit and inverter type type that makes up the solar resource site. It includes the Solar Generator Group details, Inverter Details, Planning Details and Inverter Step-Up Transformer Data. Fields include: PVGR Group #, Inverter Type, Instantaneous Controlled Fault Current Magnitude (Multiple of full load current), Controlled Fault Current Magnitude at 2 to 3 cycles after fault (Multiple of full load current and Controlled Fault Current Magnitude at 4 plus cycles after fault (Multiple of full load current)

	INVERTER DETAILS			
Skid/Array Configuration Identifier	Inverter Manufacturer	Inverter Model	MW Rating for this model of Inverter	Number of Inverters per Skid/Array Transformer
All Caps	All Caps	Numeric	MW	#
SKID_A	ACME	1001	1	10
SKID_B	BETA	9995	1	10

- **A Resource Entity may aggregate photovoltaic generators together to form a PVGR subject to provisions of Protocol Section 3.10.7.2 (9).**

- **A Resource Entity may also group PVGRs into a PVGR Group.** The group is a collection of two or more PVGRs whose performance in responding to SCED Dispatch Instructions will be assessed as an aggregate for Generation Resource Energy Deployment Performance (GREDP). A PVGR Group cannot contain any PVGRs that are Split Generation Resources. Additionally, only PVGRs that have the same Resource Node can be mapped to a PVGR Group.

Example: Non-Grouped PVGRs

For Non_Grouped PVGRs do not complete the PVGR Group #.

SOLAR GENERATION GROUP DETAILS			
Resource Name (Unit Code/Mnemonic)	PVGR Group	Group Code	SITE_GROUP
List	#	Automatic	Automatic
APOLLO_UNIT1		APOLLO_UNIT1_1	
APOLLO_UNIT2		APOLLO_UNIT2_2	

ample: Grouped PVGRs

Grouped PVGRs must complete the PVGR Group #.

SOLAR GENERATION GROUP DETAILS			
Resource Name (Unit Code/Mnemonic)	PVGR Group	Group Code	SITE_GROUP
List	#	Automatic	Automatic
APOLLO_UNIT1	1	APOLLO_UNIT1_1__1	APOLLO_1
APOLLO_UNIT2	1	APOLLO_UNIT2_2__1	APOLLO_1

7.5 Panel Details - SOLAR

The Panel Details tab identifies the different configurations of panels use at a site. These will be attached to the inverters in the Connectivity tab.

Panel Configuration Identifier	Panel Model	Panel Area	Tracking Type
All Caps	All Caps	Meters Squared	List
A	PANEL 1	50	Single axis
B	PANEL 2	60	Double axis

7.6 PVGR Connectivity

This tab allows the resource to connect the inverters and panels into units to fully describes the equipment comprising the generating untit

SOLAR GENERATION GROUP DETAILS							
Resource Name (Unit Code/Mnemonic)	PVGR Group	Group Code	SITE_GROUP	Skid/Array Configuration identifier	Number of Skid/Arrays per Skid/Array Configuration Identifier	Panel Configuration Identifier	Number of Panels per Panel Configuration
List	#	Automatic	Automatic	List	#	List	#
APOLLO_UNIT1	1	APOLLO_UNIT1_SKID_A_A_1	APOLLO_1	SKID_A	5	A	5
APOLLO_UNIT1	1	APOLLO_UNIT1_SKID_B_B_1	APOLLO_1	SKID_B	5	B	5
APOLLO_UNIT2	1	APOLLO_UNIT2_SKID_A_A_1	APOLLO_1	SKID_A	10	A	10

7.7 Ownership - RENEWABLE

The Ownership tab must be completed for each unit that is part of the Renewable resource site. It includes the Generator details, ownership information and percentages. (Fields include: Unit Name, Resource Name, Ownership start date & Ownership stop date)

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Representation of Facility Output

*This worksheet tab applies to all **WIND** Generation Resources. This tab identifies the Resource Entities that represent the output of the facility to ERCOT.*

Must complete all applicable cells in this section.

A	B	C	D	E	F	G	H	I
GENERATOR DETAILS		OWNERSHIP PERCENTAGES						
Unit Name	Resource Name (Unit Code/Mnemonic)	Joint Ownership?	Resource Entity Name	Resource Duns Number	Fixed Ownership	Master Owner	Ownership Start Date	Ownership Stop Date
List	Automatic	Y/N				Y/N	mm/dd/yyyy	mm/dd/yyyy

7.8 Parameters - RENEWABLE

The Parameters tab must be completed for each unit that is part of the Renewable resource site. It includes the Generator details, reasonability limits, seasonal ratings and unit de-rating values for temperature changes. (Fields include: Unit Name, Resource Name & Unit De-rating values). Unit De-Rating Values For Different Temperatures are the Net Max. MW rating at that temperature.

GENERATOR DETAILS		REASONABILITY LIMITS				SEASONAL RATINGS - SPRING				
Unit Name	Resource Name (Unit Code/Mnemonic)	High Reasonability Limit	Low Reasonability Limit	High Reasonability Ramp Rate Limit	Low Reasonability Ramp Rate Limit	Seasonal Net Max Sustainable Rating - Spring	Seasonal Net Min Sustainable Rating - Spring	Seasonal Net Max Emergency Rating - Spring	Seasonal Net Min Emergency Rating - Spring	Seasonal Net Max Sustainable Rating - Summer
List	Automatic	MW	MW	MW/min	MW/min	MW	MW	MW	MW	MW
UNIT1	TEST_UNIT1	75.00	0.00	15.00	1.00	75.00	0.00	75.00	0.00	75.00
UNIT2	TEST_UNIT2	40.00	0.00	5.00	1.00	40.00	0.00	40.00	0.00	40.00

7.9 Operational Parameters - RENEWABLE

The Operational Parameters tab must be completed for each unit that is part of the Renewable resource site. It includes the Generator details and the resource parameters including minimum on-line and off-line times, hot, cold, intermediate, hot to intermediate and intermediate to cold start times and daily and weekly start data. (Fields include: Unit Name & Resource Name)

GENERATOR DETAILS		RESOURCE PARAMETERS								
UNIT NAME	Resource Name (Unit Code/Mnemonic)	Minimum On Line Time	Minimum Off Line Time	Hot Start Time	Intermediate Start Time	Cold Start Time	Max Weekly Starts	Max On Line Time	Max Daily Starts	Max \ En
List	Automatic	Hours	Hours	Hours	Hours	Hours		Hours		M
UNIT1	TEST_UNIT1	0.50	1.00	0.10	0.10	0.10	85	4000.00	12	30
UNIT2	TEST_UNIT2	0.50	1.00	0.10	0.10	0.10	85	4000.00	12	16

7.10 Operational Parameters - NRRC

The Operational Parameters Normal Ramp Rate Curve data must be completed for each unit that is part of the Renewable resource site. Note that only 1 MW value and the corresponding normal upward and normal downward ramp rate are required, additional MW's are optional. (Fields include: Unit Name, Resource Name & MW number)

GENERATOR DETAILS		NORMAL RAMP RATE CURVE				
Unit Name	Resource Name (Unit Code/Mnemonic)	MW Number	NRRC Code	NRRC MW	Upward Ramprate	Downward Ramprate
List	Automatic	List	Automatic	MW	MW/min	MW/min
UNIT1	TEST_UNIT1	MW1	TEST_UNIT1_MW1_NRRC	1.00	1.00	1.00
UNIT1	TEST_UNIT1	MW2	TEST_UNIT1_MW2_NRRC	20.00	5.00	5.00
UNIT2	TEST_UNIT2	MW1	TEST_UNIT2_MW1_NRRC	1.00	1.00	1.00
UNIT2	TEST_UNIT2	MW2	TEST_UNIT2_MW2_NRRC	25.00	5.00	5.00

7.11 Operational Parameters - ERRC

The Operational Parameters Emergency Ramp Rate Curve data must be completed for each unit that is part of the Renewable resource site. Note that only 1 MW value and the corresponding normal upward and normal downward ramp rate are required, additional MW's are optional. (Fields include: Unit Name, Resource Name & MW number)

GENERATOR DETAILS		EMERGENCY RAMP RATE CURVE				
Unit Name	Resource Name (Unit Code/Mnemonic)	MW Number	ERRC Code	ERRC MW	Upward Ramprate	Downward Ramprate
List	Automatic	List	Automatic	MW	MW/min	MW/min
UNIT1	TEST_UNIT1	MW1	TEST_UNIT1_MW1_ERRC	1.00	1.00	1.00
UNIT1	TEST_UNIT1	MW2	TEST_UNIT1_MW2_ERRC	20.00	10.00	10.00
UNIT2	TEST_UNIT2	MW1	TEST_UNIT2_MW1_ERRC	1.00	1.00	1.00
UNIT2	TEST_UNIT2	MW2	TEST_UNIT2_MW2_ERRC	25.00	10.00	10.00

7.12 Private Network - Unit

The Private Network Unit tab is to be completed only if the site has completed the site information as a PUN site, then the data must be completed for each unit that is part of the private network resource site, Which includes the MW and MVAR load, net interchange and gross unit capabilities. If the site is not part of a PUN, then no unit data need be filled out on this tab and all fields will be blacked out. (Fields include: Unit Name, Site Code & Resource Name)

GENERATOR DETAILS			PRIVATE NETWORK - UNIT INFORMATION				
Unit Name	SITE CODE	Resource Name (Unit Code/Mnemonic)	Average Amount Of Self-serve Private Load	Average Amount Of Self-serve Private Reactive Load	Expected Typical Private Network Net Interchange	Expected Typical Private Network Net Reactive Interchange	Private Network Gross Unit Capability
List	Automatic	Automatic	MW	MVAR	MW	MVAR	MW

7.13 Reactive Capability - RENEWABLE

The Reactive Capability must be completed for each unit that is part of the Renewable resource site. It includes the generator details and the reactive capability curve data which are 5 increasing MW values of Operating Real Power and 9 points of MVAR reactive power information. The lagging MVAR amounts should be listed as positive numbers and the leading MVAR amounts should be listed as negative. MW5 is the unity power factor amount or maximum value of real power MW at 1.0 PF, the reactive power which crosses the x-axis on the curve is zero. (Fields include: Unit Name, Resource Name, Gross/Net values, NDCRC test & Test date)

GENERATOR DETAILS	REACTIVE CAPABILITY CURV								
Resource Name (Unit Code/Mnemonic)	Reactive Capability Data Provided is Gross/Net Values?	Reactive Capability Data Provided is from NDCRC Test Data?	If Reactive Capability Data Provided Is From NDCRC test Data Then Enter The Date On Which The Test Was Performed.	Mw1	Lagging MVAR Limit Associated With Mw1 Output	Leading MVAR Limit Associated With Mw1 Output	Mw2	Lagging MVAR Limit Associated With Mw2 Output	Leading MVAR Limit Associated With Mw2 Output
Automatic	List	Y/N	mm/dd/yyyy	MW	MVAR	MVAR	MW	MVAR	MVAR
TEST_UNIT1	NET	Y	01/01/2011	0.50	0.51	-0.51	5.50	2.40	-1.80
TEST_UNIT2	NET	Y	01/01/2011	1.00	0.20	-0.21	2.90	1.10	-1.00

7.14 Planning - RENEWABLE

The Planning tab must be completed for each unit that is part of the Renewable resource site. It includes the generator auxiliary load information, generation auxiliary MW load characteristics and generation auxiliary MVAR load characteristics.

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Planning Information

This worksheet tab provides ***aggregated*** *planning information for* ***WIND*** *generation resources (WGR) as listed in Unit Info - Wind tab using turbine type and number from Unit Info - Wind Turbine tab and parameters fr*

Must complete all applicable cells in this section.

	GENERATOR AUXILIARY LOAD			GENERATION AUXILIARY LOAD CHARACTERISTICS FOR MW LOAD					GENERATION AUXILIARY LOAD CHARACTERISTICS FOR MVAR LOAD			
Resource Name (Unit Code/Mnemonic)	Average Amount Of Auxiliary Real Power	Average Amount Of Auxiliary Reactive Power	Auxiliary Load Power Factor	Large Motor, Percent Of Total Mw Load	Small Motor, Percent Of Total Mw Load	Resistive (heating) Load, Percent Of Total Mw Load	Discharge Lighting, Percent Of Total Mw Load	Other, Percent Of Total Mw Load	Large Motor, Percent Of Total MVAR Load	Small Motor, Percent Of Total MVAR Load	Discharge Lighting, Percent Of Total MVAR Load	Other, Percent Of Total MVAR Load
List	MW	MVAR										

7.15 Protection - RENEWABLE

The Protection tab must be completed for each unit that is part of the Renewable resource site. It includes the generator details, plant under and over voltage protection and plant under and over frequency protection information. (Fields include: Unit Name, Site Code & Resource Name)

GENERATOR DETAILS										
Unit Name	Site Code	Resource Name (unit Code/mnemonic)	Breaker Interruption Time	Instantaneous Undervoltage Trip	Undervoltage 1	Time 1	Undervoltage 2	Time 2	Undervoltage 3
List	Automatic	Automatic	cycles	p.u.	p.u.	sec	p.u.	sec	p.u.
UNIT1	TEST	TEST_UNIT1	1		1.10000	0.74	0.20000	0.69	0.10000
UNIT2	TEST	TEST_UNIT2	1		0.20000	0.01	0.20000	0.69	0.10000

7.16 Collector System - RENEWABLE

The Collector System tab must be completed for each unit that is part of the Renewable resource site. It includes the system details for cable types, the positive sequence data and zero sequence data. Also attach either the collection system one-line or cable segment model data in fields provided. (all are new fields) See section 3.2 for process to embed files.

COLLECTION SYSTEM ONE LINE DIAGRAM (Embed a PDF One Line Diagram)

Adobe Acrobat Document

COLLECTION SYSTEM DETAILED MODEL (Embed a PSS/e Raw Data or Powerwold Model)

COLLECTION SYSTEM DETAILED MODEL		POSITIVE SEQUENCE DATA			ZERO SEQUENCE DATA	
Cable Type	Voltage Level kV	R/kft	X/kft	Charging Bc/kft	R0/kft	X0/kft
	kV	(p.u. on 100 MVA base)	(p.u. on 100 MVA base)	(p.u. on 100 MVA base)	(p.u. on 100 MVA base)	(p.u. on 100 MVA ba
GE 1.5	35.00	3.50000	0.35000	0.03500	0.35000	0.03500
GE 1.5	35.00	3.50000	0.35000	0.03500	0.35000	0.03500
V90	25.00	2.50000	0.25000	0.02500	0.25000	0.02500
V90	25.00	2.50000	0.25000	0.02500	0.25000	0.02500

7.17 Collector System – Renew Segment data

The Collector System tab segment data must be completed for each cable type that is part of the Renewable resource site. It includes the cable type and the segment details. (all are new fields)

Cable Type	WIND SEGAMENT DETAILS					
Cable Type	From	To	Circuit Number	Voltage Level	Cable Segment Length	Number of Turbines On Cable Segment
List	Integer	Integer	Integer	kV	in kft	Integer
GE 1.5	5.00	10.00	5.00	34.50	2.00	5.00
GE 1.5	5.00	10.00	5.00	34.50	2.00	5.00
GE 1.5	5.00	10.00	5.00	34.50	5.00	10.00
GE 1.5	20.00	50.00	5.00	34.50	5.00	10.00
GE 1.5	20.00	50.00	5.00	34.50	5.00	20.00
GE 1.5	20.00	50.00	5.00	34.50	5.00	20.00
V90	1.00	10.00	10.00	34.50	1.00	2.00
V90	1.00	10.00	10.00	34.50	1.00	2.00
V90	5.00	20.00	10.00	34.50	2.00	8.00
V90	5.00	20.00	10.00	34.50	2.00	8.00
V90	10.00	50.00	10.00	34.50	4.00	10.00
V90	10.00	50.00	10.00	34.50	4.00	10.00

Embed the stability study model to this tab. See section 3.2 for process.

7.18 PSCAD Model

Embed PSCAD model to this tab. See section 3.2 for process.

7.19 Dynamic Data

Embed Dynamic Data model to this tab. See section 3.2 for process.

8.0 LaaR RARF Workbook

8.1 Instructions

Detailed instructions for the RARF submittal process through the Market Information System (MIS) are contained in section 3.1 of this document.

8.2 General Information - ALL

The General Information tab identifies information about the Resource Entity. The contact information is essential, as it provides ERCOT with additional contacts in case of questions regarding the RARF. Enter the Resource Entity Name that matches the name ERCOT has on record, note that it does not allow any special characters except spaces and dashes. The Data Universal Numbering System (DUNS) number is a 9 or 13 digit number listed with ERCOT to uniquely identify an entity.

This submittal is for:	Revisions
* Deletions are accepted as intentions. This form does not supersede the Notice of Suspension of Operations requirements.	
Submittal Information	
Date Form Completed:	05/05/2010
Resource Entity Submitting Form:	RE NAME HERE
Resource Entity DUNS #:	1234567891000
Primary Contact (name of person ERCOT can contact with questions regarding this form)	
Printed Name:	JOHN DOE
Title:	MGR OPERATIONS
Phone Number:	555-123-4567
E-mail Address:	jdoe@email.com
Fax Number:	
Secondary Contact (if available)	
Printed Name:	SAM WORKER
Title:	SUPERVISOR OPERATIONS
Phone Number:	512-123-9876
E-mail Address:	sworker@email.com
Fax Number:	

8.3 Load Resource Information

The Load Resource information tab includes the unit details for each of the loads acting as a resource. There are two types of load resources, controllable load resource and non-controllable load resource. Each load resource must be individually registered with ERCOT by ESI-ID and are subject to qualification testing.

Common Name for Load Resource	Dispatch Asset Code (provided by ERCOT)	Physical Street Address for Point of Delivery (POD)	Name of City for Point of Delivery (POD)	Is Load Netted From Generation at ERCOT Read Gensite?	Is Load Behind a NOIE Settlement Meter Point?	Load Resource Type (CLR/UFR/Interruptible)	If CLR, Dyn Sch
				Y/N	Y/N		
My Load Resource #1	MY_LD1	123 First Street	Township	N	N	UFR	
My Load Resource #2	MY_LD2	123 First Street	Township	N	N	UFR	

8.4 Load Resource Parameters

This section must be completed for each dispatch code that is part of a load resource. It includes the Non-CLR resource parameters, the CLR resource parameters, CLR normal ramp rate curve data and CLR emergency ramp rate curve data.

	Non-CLR Resource Parameters							CLR R
	Minimum Interruption Time	Minimum Restoration Time	Max WEEKLY Deployments	Max Interruption Time	Max DAILY Deployments	Max Weekly Energy	Minimum Notice Time	Max Deplo Tim
Dispatch Asset Code	hours	hours		hours		MWh	minutes	hour
MY_LD1	0	1	1	4	1	15.00	10	
MY_LD2	0	1	1	4	1	15.00	10	

8.5 One Line

Embed the one line diagram(s) in pdf or cad to this tab. See section 3.2 for process.

9.0 Transmission RARF Workbook

9.1 Instructions

Detailed instructions for the RARF submittal process through the Market Information System (MIS) are contained in section 3.1 of this document.

9.2 Station

Thae Station data tab is to be completed for each station that the RE owns. There should be one line of data entered for each voltage contained in the station. The Station tab contains data describing the resource owned station.

9.3 Line Data

The Line data tab is to be completed for each line that is located at the resource site and consists of the Line details, the “to” connected station and devices and the “from” connected station and devices. (Fields include: Zero Sequence Line Resistance, Zero Sequence Line Reactance, Zero Sequence Charging Susceptance & Line Code). Note: Cell B5 (Resource Site Code) must be completed on all forms.

A	B	C	D	E	F	G	H	I
ERCOT Confidential								
Line Data								
Please submit all Station's one-lines which are affected by the line submittal.								
Resource Site Code:								
	Resource Site Code must be all CAPS.	LINE (ANY LENGTH) DETAILS						
DESCRIPTION OF CHANGE	Line Name	Line Voltage Level	Resistance In P.u. (100 Mva Base)	Reactance In P.u. (100 Mva Base)	Charging Susceptance In P.u. (100 Mva Base)	Zero Sequence Line Resistance In P.u. (100mva Base)	Zero Sequence Line Reactance In P.u. (100mva Base)	Zero Sequen Charging Susceptance P.u. (100mv Base)
List	enter all caps	kV	p.u.	p.u.	p.u.	p.u.	p.u.	p.u.

Note: Prior to your next RARF submission, remove “Add” or “Change” from the Description of Change column for all equipment that is not being updated on this submission. Only use “Add” or “Change” for new updates on this submission. If equipment was “Deleted” on prior submission, remove all data from that row before the next submission.

9.4 Line Temperature

The Line temperature tab is to be completed for each line that is located at the resource site and consists of the Line details, noting whether the line is static or dynamic in nature. If static then only the Nominal ratings would be completed, if dynamic then all the ratings including temperatures from 20 to 115 degrees need to be completed. (Fields include: Line Code & Planning Rate C ratings)

			Nominal				20 °F			
Line Name	Line Code	Line Rating	Continuous Rating	2-hr Emergency Rating	15-min Rating	Planning Rate C	Continuous Rating	2-hr Emergency Rating	15-min Rating	Planning Rate C
Automatic	Automatic	Static/Dynamic	MVA	MVA	MVA	MVA	MVA	MVA	MVA	MVA
TEST_LINE	TEST_TEST_TEST_LINE	Static	968.00	1076.00	1076.00	968.00	[redacted]			
TEST_TIE	TEST_TEST_TEST_TIE	Static	159.00	159.00	173.00	159.00				

9.5 Breaker Switch Data

The Breaker Switch data tab is to be completed for each breaker or switch that is located at the resource site. It consists of the breaker or switch details, the static ratings and up to 10 connected devices for side 1 and 10 connected devices for side 2. (Fields include: Switch Code & Static ratings)

	BREAKER / SWITCH						Static Ratings for Planning				
DESCRIPTION OF CHANGE	Switch Name	Ercot Station Code Mnemonic	Switch Code	Is This A Fault Isolating Device (e.g. Circuit Breaker)	Normal Operating Status (when In-service)	Voltage Level	Continuous Rating	2-hr Emergency Rating	15 Min Rating	Connected Device 1	Connected Device 2
List	enter all caps	enter all caps	Automatic	Y/N	Open/Closed	kV	MVA	MVA	MVA	enter all caps	enter all caps
ADD	DSC1	TEST	TEST_TEST_DSC1	Y	CLOSED	69.00	968.00	1076.00	1076.00	TEST_LINE	DSC2
ADD	CB1	TEST	TEST_TEST_CB1	N	CLOSED	69.00	159.00	159.00	173.00	TEST_LINE	CB1
ADD	CB2	TEST	TEST_TEST_CB2	N	CLOSED	69.00	159.00	159.00	173.00	TEST_LINE	CB2
ADD	DSC10	TEST	TEST_TEST_DSC10	Y	CLOSED	69.00	159.00	159.00	173.00	TEST_TIE	DSC11
ADD	DSC11	TEST	TEST_TEST_DSC11	Y	CLOSED	69.00	159.00	159.00	173.00	TEST_TIE	DSC10

Note: Prior to your next RARF submission, remove "Add" or "Change" from the Description of Change column for all equipment that is not being updated on this submission. Only use "Add" or "Change" for new updates on this submission. If equipment was "Deleted" on prior submission, remove all data from that row before the next submission.

9.6 Capacitor and Reactor Data

The Capacitor and Reactor data tab is to be completed for each capacitor and reactor that is located at the resource site. It consists of the capacitor and reactor details and up to 10 connected devices. (Fields include: Device Code & Zero Sequence Reactance)

DESCRIPTION OF CHANGE	Device Name	Ercot Station Code Mnemonic	Device Code	Capacitor Or Reactor	Nominal Mvar	Voltage Level Kv	Bus Number (PTI Bus Number)	Automatic Voltage Regulation	Voltage Level Of Busbar Being Regulated	Desired Regulating Voltage	Minimum Regulating Voltage
List	enter all caps	enter all caps	Automatic	List	MVAR	kV	#	Y/N	kV	kV	kV
ADD	C1	TEST	TEST_TEST_C1	C	17.50	34.50	8523	Y	69.00	69.00	66.00
ADD	C2	TEST	TEST_TEST_C2	C	17.50	34.50	9632	Y	69.00	69.00	66.00

Note: Prior to your next RARF submission, remove "Add" or "Change" from the Description of Change column for all equipment that is not being updated on this submission. Only use "Add" or "Change" for new updates on this submission. If equipment was "Deleted" on prior submission, remove all data from that row before the next submission.

9.7 Transformer Data

The Transformer data tab is to be completed for each transformer that is located at the resource site. It consists of the transformer details, units associated if it is a generator step up transformer, high side voltage information, low side voltage information, voltage regulation, low tap settings and high tap settings. (Fields include: Transformer Code, Zero Sequence Data, Zero Sequence Grounding Resistance, Zero Sequence Grounding Reactance, Zero Sequence Resistance, Zero Sequence Reactance, Positive Sequence Resistance & Positive Sequence Reactance)

DESCRIPTION OF CHANGE	Transformer Name	Ercot Station Code Mnemonic	Transformer Code	Transformer Test Report Attached?	Transformer In a Master-follower Current Balancing Configuration?	Master Name (can Be Same As this transformer)	Follower Name (can Be Same As this transformer)	Generator Step Up Transformer?	Zero Sequence Data - Winding Connection Code (1-8)
List	enter all caps	enter all caps	Automatic	Y/N	Y/N	enter all caps	enter all caps	Y/N	p.u.
ADD	FMR1	TEST	TEST_TEST_FMR1	Y	N			Y	2.00000
ADD	XF2	TEST	TEST_TEST_XF2	Y	N			Y	2.00000

Note: Prior to your next RARF submission, remove "Add" or "Change" from the Description of Change column for all equipment that is not being updated on this submission. Only use "Add" or "Change" for new updates on this submission. If equipment was "Deleted" on prior submission, remove all data from that row before the next submission.

9.8 Transformer Tap settings

The Transformer Tap Settings tab is to be completed for each transformer that is located at the resource site. It consists of the Description of Change Transformer Name, ERCOT Station Code Mnemonic, Transformer Code High or Low side connection flag and the Tap Setting positions and voltage levels.

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Transformer Data

Please submit one-line diagram together with this Transformer data request form.

Must complete all applicable cells in this section.

DESCRIPTION OF CHANGE	Transformer Name	Ercot Station Code Mnemonic	Transformer Code	High-Low Flag	Tap Position At Manufactured Nominal Voltage	Total Number Of Tap Positions	No F
List	List	List	Automatic	List	#	#	

Note: Prior to your next RARF submission, remove "Add" or "Change" from the Description of Change column for all equipment that is not being updated on this submission. Only use "Add" or "Change" for new updates on this submission. If equipment was "Deleted" on prior submission, remove all data from that row before the next submission.

9.9 Static Var Compensator Data

The Static Var Compensator data tab is to be completed for each svc that is located at the resource site. It consists of the svc details and up to 10 connected devices. (Fields include: SVC Code, Fixed MVAR, Minimum Admittance Limits, Maximum Admittance Limits, Minimum Threshold, Maximum Threshold, Minimum Voltage Threshold & Maximum Voltage Threshold)

DESCRIPTION OF CHANGE	SVC Name	Ercot Station Code Mnemonic	SVC Code	Svc Base Voltage Level	Fixed Mvar (var Injection At Nominal Voltage)	Minimum Admittance Limits (100 Mva Base)	Maximum Admittance Limits (100 Mva Base)	Minimum Steady State Reactive Power Limits	Maximum Steady State Reactive Power Limits
List	enter all caps	enter all caps	Automatic	kV	MVAR	p.u.	p.u.	MVAR	MVAR
ADD	DST1	TEST	TEST_TEST_DST1	34.50	7.50	7.50000	7.50000	7.50	7.50
ADD	DST2	TEST	TEST_TEST_DST2	34.50	7.50	7.50000	7.50000	7.50	7.00

Note: Prior to your next RARF submission, remove "Add" or "Change" from the Description of Change column for all equipment that is not being updated on this submission. Only use "Add" or "Change" for new updates on this submission. If equipment was "Deleted" on prior submission, remove all data from that row before the next submission.

9.10 Series Device Data

The Series Device data tab is to be completed for each series device that is located at the resource site. It consists of the series device details and up to 10 connected devices for side 1 and 10 connected devices for side 2. Series devices are impedance element which may be series reactors and capacitors or may be short lines within a station. (Fields include: SD Code)

	SERIES DEVICE DATA									
DESCRIPTION OF CHANGE	Series Device Name	Ercot Station Code Mnemonic	SD Code	Voltage Level	Resistance	Reactance	Continuous Rating	2-hr Emergency Rating	15-min Rating	D
List	enter all caps	enter all caps	Automatic	kV	p.u.	p.u.	MVA	MVA	MVA	ent
ADD	LN_1	TEST	TEST_TEST_LN_1	69.00	0.00480	0.02510	167.00	185.00	185.00	DS
ADD	LN_2	TEST	TEST_TEST_LN_2	69.00	0.00110	0.00540	167.00	185.00	185.00	DS

Note: Prior to your next RARF submission, remove "Add" or "Change" from the Description of Change column for all equipment that is not being updated on this submission. Only use "Add" or "Change" for new updates on this submission. If equipment was "Deleted" on prior submission, remove all data from that row before the next submission.

9.11 Load Data

The Load data tab is to be completed for each load that is located at the resource site. It consists of the load details and up to 10 connected devices. For the transmission grid (greater than 60kV) and for the kV grids modeled for output paths of generators (less than 60kV), all auxillary, station service and/or power transformers are modeled in ERCOT as a load point at the high side of these types of transformers. (FieldsFields include: Load Code)

DESCRIPTION OF CHANGE	LOAD DATA								
	Ercot Station Code Mnemonic	Load Name	Load Code	Load Voltage Level	Ptl Bus Number	Average Load Under Normal Operations	Average Mvar Under Normal Operations	Directly Connected Device 1	Directl Connect Device
List	enter all caps	enter all caps	Automatic	kV	#	MW	MVAR	enter all caps	enter all
ADD	TEST	LD1	TEST_TEST_LD1	34.00	1235	14.00	10.00	DSC123	CB100
ADD	TEST	AUX1	TEST_TEST_AUX1	34.00	1236	8.00	6.00	DSC125	CB200

Note: Prior to your next RARF submission, remove "Add" or "Change" from the Description of Change column for all equipment that is not being updated on this submission. Only use "Add" or "Change" for new updates on this submission. If equipment was "Deleted" on prior submission, remove all data from that row before the next submission.

9.12 PUN Load Data

The PUN Load data tab is to be completed for each load that is located at Private Use Network (PUN) resource site. It consists of the station details for each load and the 168 hour (24 hours and 7 days) MW data for each load. This tab must be manually populated for the 5.1 version

Station Details			168 HR Data		
Station Code	Load Name	Load Code	Hour Ending	Day of the Week	MW
enter all caps	List	Automatic	Automatic	Automatic	
TEST	LD1	TEST_TEST_LD1	1	Monday	2.0
[redacted]			2	Monday	2.0
			3	Monday	2.0
			4	Monday	2.0
			5	Monday	2.0
			6	Monday	2.0
			7	Monday	10.0
			8	Monday	10.0
			9	Monday	14.0
			10	Monday	14.0
			11	Monday	14.0
			12	Monday	14.0
			13	Monday	14.0
			14	Monday	14.0
			15	Monday	14.0
			16	Monday	14.0
			17	Monday	14.0
			18	Monday	14.0
			19	Monday	10.0
			20	Monday	10.0
			21	Monday	2.0
			22	Monday	2.0
			23	Monday	2.0
			24	Monday	2.0
			1	Tuesday	2.0
			2	Tuesday	2.0
			3	Tuesday	2.0
			4	Tuesday	2.0
			5	Tuesday	2.0
			6	Tuesday	2.0
			7	Tuesday	10.0

9.13 One Line

Embed the one line diagram(s) in pdf or cad to this tab. See section 3.2 for process.

9.14 Transformer Test Data

All One Lines will have to be Reembeded
Embed the transformer test data to this tab. See section 3.2 for process.

10.0 Settlement Only Distributed Generation (SODG)

10.1 Instructions

Detailed instructions for the RARF submittal process through the Market Information System (MIS) are contained in section 3.1 of this document.

10.2 Unit Info – DG

This section must be completed for each Settlement Only Distributed Generation (SODG) unit. It includes the Generator Details, Unit Dates, Fuel Information, Private Network Info, Generic Info, Mapping Instructions and Resource Owner Data. Cell B5 (Resource Site Code) must be completed on all forms.

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Unit Information - Settlement Only Distributed Generator

This worksheet tab provides generator Unit information for Distributed and Non-Modeled Generator assets.

Please complete this section.

Resource Site Code:

Generation Details		Unit Dates		Fuel Information					
Unit Name	Unit Code	Unit Start Date	Unit End Date	Name Plate Rating	Technology Type	Name Plate Rating	Renewable	Renewable/ Offset	If Wind, Number of Turbines
all caps	Automatic	mm/dd/yyyy	mm/dd/yyyy	MW	List	MVA	Y/N	List	#